Mohamed Eljahmi
RBE-550: Motion Planning
Homework #1
September 01, 2025

References

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- Hwang, M., & Rucker, D. C. (2021). "Concentric Tube Robots in Surgery: Current Applications and Future Directions." *Annual Review of Control, Robotics, and Autonomous Systems*.
- Yan, P., et al. (2019). "Surgical Robot Motion Planning Under Uncertainty." *IEEE Transactions on Medical Robotics and Bionics*, 1(2).
- https://docs.python.org/3/tutorial/venv.html
- https://openai.com/index/chatgpt/
- https://ubuntu.com/download/desktop
- https://stackoverflow.com/
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Git repository
https://github.com/meljahmi-personal/RBE550-assignment1.git

Introduction:
My name is Mohamed Eljahmi. I earned a BSEE from Northeastern University (1983), an MSc in Electrical & Computer Engineering from WPI (2023), and I am currently pursuing an MSc in Robotics Engineering at WPI, with graduation planned for May 2026.
Throughout my career, I have worked in various industries, including embedded systems, telecommunications, medical devices, consumer electronics, analytics, defense, aerospace, and finance. While my current role is in software engineering within the financial sector, I am actively seeking a position in robotics and embedded systems.

I am actively investing in my education, financing two graduate degrees at WPI, and I have developed strong interests in machine learning, AI, robotics, and autonomous systems. I also plan to pursue side projects with robotics kits and drones to improve my skills in obstacle avoidance, motion planning, and robotic system integration.

Motion Planning Application

Motion Planning in Surgical Robots

The Problem

- In minimally invasive surgery (laparoscopy, robotic-assisted surgery), the robot must maneuver instruments through tiny incisions and around sensitive anatomy.
- The workspace is crowded with blood vessels, nerves, and organs.
- The planner must generate collision-free paths for tools, often in constrained, deformable environments.

Why Motion Planning is Needed

- Collision avoidance: A wrong move could puncture a vessel or damage nerves.
- Constrained kinematics: Instruments pivot at the incision point (called the "remote center of motion"), which restricts motion.
- Precision: Motions must be extremely fine-grained (millimeters).
- Dynamic adaptation: Organs can move or deform due to breathing, heartbeat, or surgical manipulation.

Planning Methods

- Sampling-based planners (RRT, PRM) are used to explore feasible paths within narrow anatomical passages.
- Optimization-based planners: refine paths to minimize tool motion, reduce trauma, or maintain visibility of the surgical field.
- Image-guided planning: combine pre-operative imaging (CT, MRI) with intra-operative updates to plan safe trajectories.
- Deformable environment models: Advanced planners account for tissue deformation and update the plan in real-time.

Applications

- Da Vinci Surgical System (Intuitive Surgical): currently widely used; path planning ensures tools respect the incision's pivot point.
- Orthopedic surgery robots (Mako, ROSA): motion planning guides tools when shaping bone for implants.
- Neurosurgery: planners ensure tools follow safe trajectories through the brain to avoid critical areas.
- Radiation therapy: planners determine safe beam paths to a tumor while avoiding healthy tissue.

Why It's Interesting

- It combines robotics, medicine, and safety-critical AI.
- Unlike cars, surgical robots must operate with zero tolerance for error — you can pull over a car, but not a patient mid-surgery.
- Motion planning has a direct impact on patient outcomes, resulting in less tissue damage, shorter recovery times, and improved surgical precision.